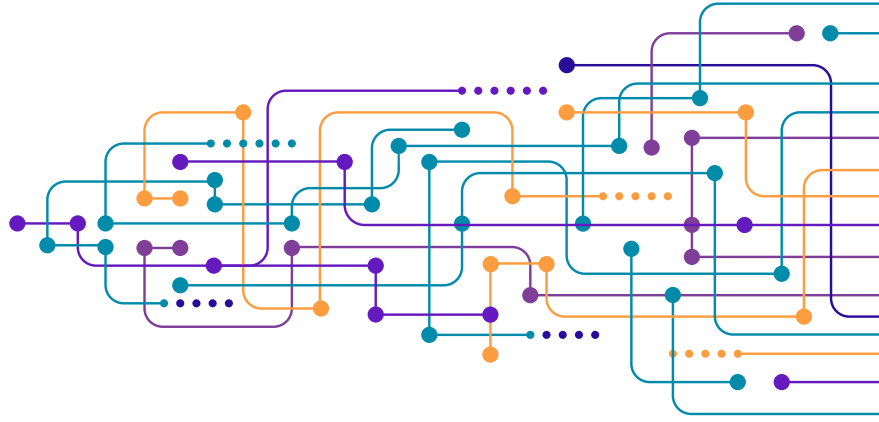


02

Role change: The AI impact



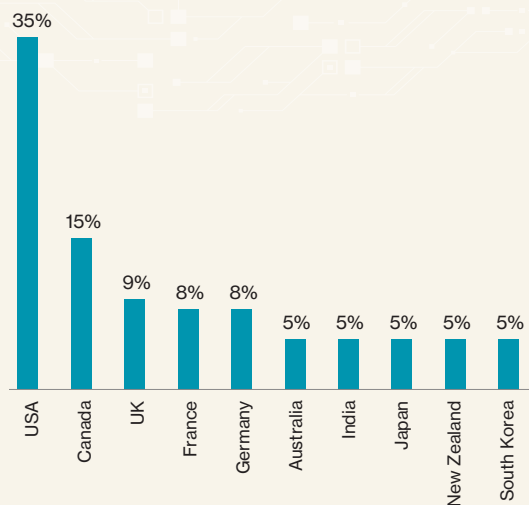
Data engineers have worked behind the scenes for years but are now capturing the attention of their organizations' senior leaders. This is largely thanks to AI. Of the senior technology leaders in our survey, 72% deem data engineers to be integral to the overall success of their business (see Figure 2a). This figure rises to 86% of respondents in the survey's largest organizations – those with annual revenues of more than \$10 billion (see Figure 2b). Sector-wise, executives in financial services and manufacturing are particularly convinced of the importance of data engineers.

According to Ritu Jyoti, formerly a senior executive at International Data Corporation (IDC) and now chief executive officer of a stealth AI startup, the explanation for data engineers' elevated profile is straightforward: "AI is increasingly a foundation of business success, and there is no AI without data."

Chris Child, vice president of product, data engineering at Snowflake, makes a similar point: "To realize AI's full potential, a strong data foundation isn't optional – it's mission critical. And we're seeing the workforce evolve to follow suit, with data engineers being viewed less

Figure 1a: Survey respondents by country (where their organization is headquartered)

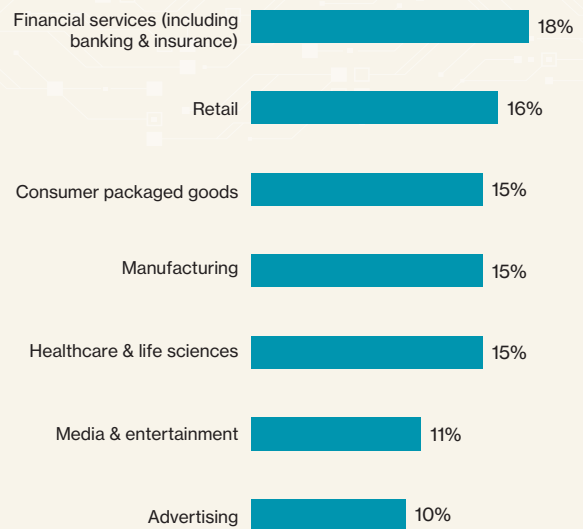
(% of respondents)



Source: MIT Technology Review Insights survey, 2025

Figure 1b: Survey respondents by primary industry

(% of respondents)



Source: MIT Technology Review Insights survey, 2025

“To realize AI’s full potential, a strong data foundation isn’t optional – it’s mission critical. If your C-suite still considers data engineering as a support role, you’re already five years behind – and probably training your future competitors.”

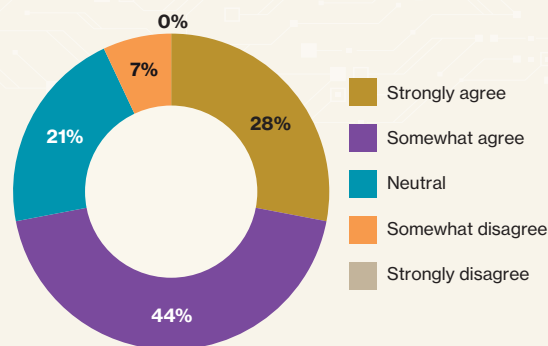
Chris Child, Vice President of Product, Data Engineering, Snowflake

as a backend plumbing function, but instead as an integral, strategic business partner responsible for enterprises’ most valuable asset: their data. If your C-suite still considers data engineering as a support role, you’re already five years behind – and probably training your future competitors.”

For the survey’s biggest companies, which placed even greater emphasis on the importance of data engineers, this is likely due to their more advanced AI maturity. Considerably more of the biggest companies have already deployed generative, multi-modal, and semi-autonomous agentic AI, as well as AI-powered data tools. “Some of those early AI adopters – the organizations that are really moving the needle on AI – are looking at their data engineers as strategic enablers in business transformation,” says Jyoti.

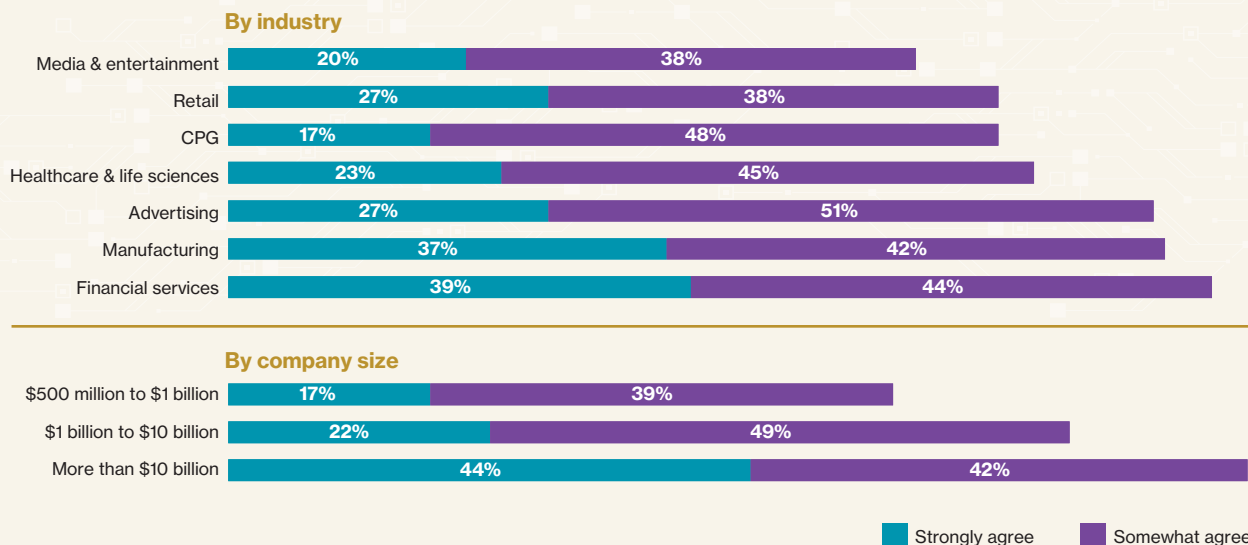
Figure 2a: Nearly 3 in 4 data and technology executives agree data engineers are integral to the success of their business

(% of respondents)



Source: MIT Technology Review Insights survey, 2025

Figure 2b: Nearly 9 in 10 of the largest surveyed organizations say data engineers are integral to the success of their business



Source: MIT Technology Review Insights survey, 2025

Growing workloads and a shifting focus

A large majority of surveyed executives (81%) say the job description for data engineers in their organization has changed radically due to AI. This can be seen in a shift in their day-to-day activities. Two years ago, data engineers spent an average of 19% of their workday on AI projects (such as building and monitoring model pipelines or data cleansing), say respondents, and 81% on other activities (data warehousing, quality checking, and infrastructure management, for example). In 2025, the analogous figures are 37% of time spent on AI and 63% on other activities (see Figure 3).

“The role has evolved significantly,” says Jyoti. “In the earlier days of AI and machine learning, data engineers focused on building data pipelines. They were very siloed, focusing on one task at a time. Now they’re focused on designing and architecting reusable data platforms, governance, and oversight. They’re playing a pivotal role as key enablers of AI.”

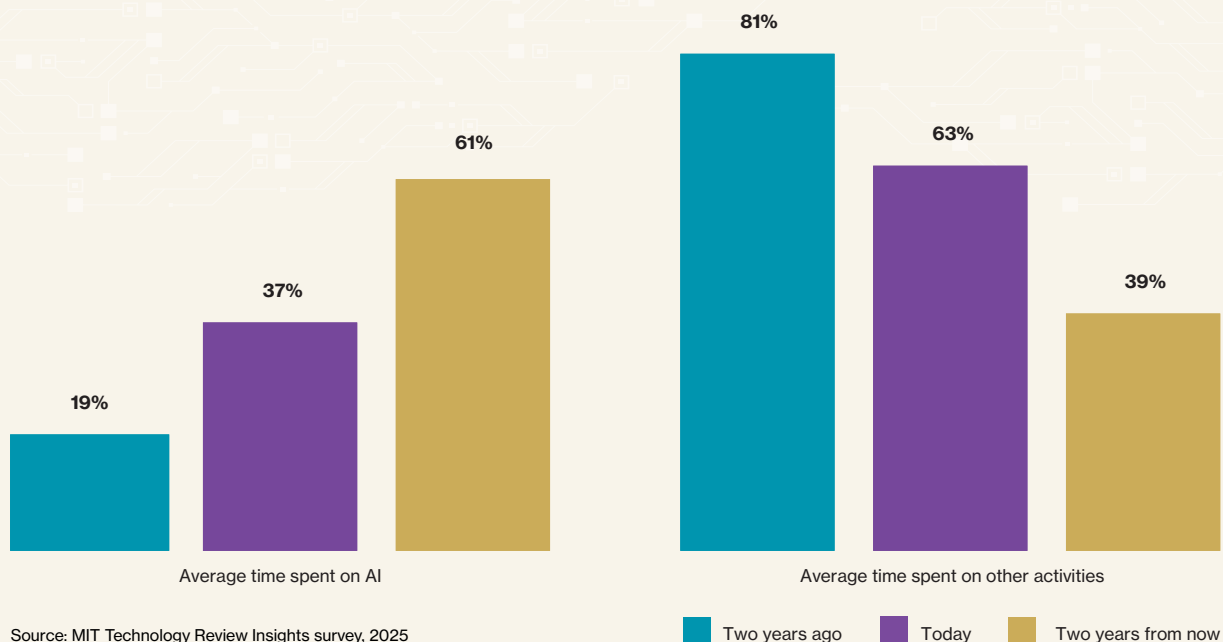
The respondents expect the change to accelerate: two years from now, they believe the ratio of time spent on AI to that on other activities will almost be the reverse of today – 61% to 39% in favor of AI (see Figure 3). As priorities change, so too does the volume of work.

“Some of those early AI adopters – the organizations that are really moving the needle on AI – are looking at their data engineers as strategic enablers in business transformation.”

Ritu Jyoti, Chief Executive Officer,
Stealth AI Startup

Most respondents (77%) say their data engineers’ workload is becoming increasingly heavy (see Figure 4a). “They’re struggling to keep up,” says Jyoti. “Data volumes and velocity have exploded. Data is now multi-modal and is being ingested in real time.” Interestingly, considerably more chief AI officers observe this than their peers (see Figure 4b).

Figure 3: Data engineers are spending increasing amounts of time on AI projects versus other activities (% of respondents)



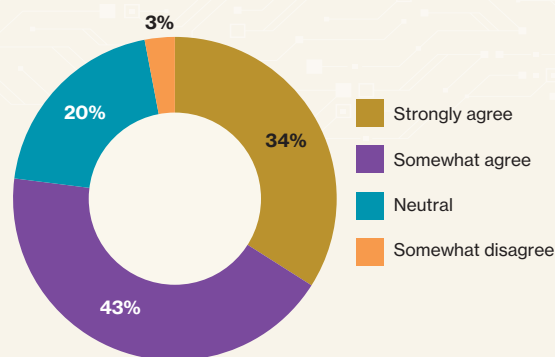
Source: MIT Technology Review Insights survey, 2025

“Models need so much more data and in multiple formats. Where it used to be making sense of structured data, which was relatively straightforward, now it’s: ‘What do we do with all this unstructured data? How do we tag it? How do we organize it? How do we store it?’ That’s a bigger challenge.”

George Westerman, Senior Lecturer and Principal Research Scientist, MIT Sloan School of Management

Figure 4a: 3 in 4 respondents agree data engineers’ workloads are growing

(% of respondents)



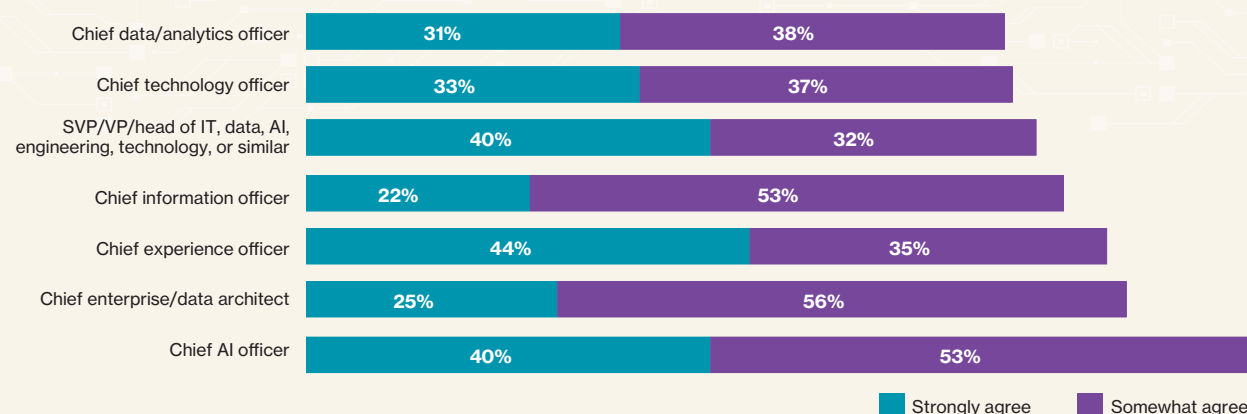
Source: MIT Technology Review Insights survey, 2025

“Models need so much more data and in multiple formats,” says George Westerman, senior lecturer and principal research scientist at the MIT Sloan School of Management. “Where it used to be making sense of structured data, which was relatively straightforward, now it’s: ‘What do we do with all this unstructured data? How do we tag it? How do we organize it? How do we store it?’ That’s a bigger challenge.”

Dave Masino, senior director of data and intelligence at technology consulting company Slalom, sees the growing workload as an opportunity. “AI has increased the amount of work data engineers are doing, but you inherently want your team to be busy,” he says. “I see the advantage we’re getting from AI-enabled acceleration eclipsing the amount of additional workload.”

Figure 4b: CAIOs are the most likely to agree that data engineers’ workloads are becoming increasingly heavy

(By executive role)



Source: MIT Technology Review Insights survey, 2025

Learning from software engineers

Data engineers' success in mastering AI can in large part be credited to their ability to adapt software engineering practices to their own needs, says Chris Child, vice president of product and data engineering at Snowflake. "To my mind this is the biggest shift in data engineering in the past five to seven years," he says, "and it's still ongoing. It's helped data engineers understand how to build and manage complex software projects, including AI projects, at scale."

Child provides examples: "It's how to treat infrastructure as code, using tools like Terraform to set up your data engineering infrastructure.

It's making sure all your data engineering code is checked into version control deployed through continuous integration and continuous deployment." This is helping with managing AI projects, Child says, because many AI tools in use today have been trained on software engineering problems.

"It's been a massive positive," says Child. "Data engineers are now thinking in a software kind of way. They're now better able to move quickly and to change things without fear of breaking something," he says. "It's been a change in philosophy and team culture as much as a change in tooling."

"AI has increased the amount of work data engineers are doing, but you inherently want your team to be busy. I see the advantage we're getting from AI-enabled acceleration eclipsing the amount of additional workload."

Dave Masino, Senior Director of Data and Intelligence, Slalom

